Conditional Probability & Bayes Classification:

For: Bayes Theorem

Conditional Prob: $P(A/B) = \frac{P(A \cdot B)}{P(B)}$ ⇒ Prob of event A when B has already occurred.

Classification: Let $x = \{x_1, x_2, \dots x_n\}$ for a sample.

↓

Class label C should be found. [$P(C/x)$].

Posterior probability ⇒ $\boxed{P(C/x) = \frac{P(C) \cdot P(x/C)}{P(x)}}$ → class label associated with x.

⇒ with the prob of feature from training dataset, class label prob is calculated.

↓ Bayes theorem

As it is the prob of class label C, after observing class label x

$\boxed{\Rightarrow P(C) \rightarrow \text{Prior probability}}$

$P(x/C)$ ⇒ prob of observing input feature x in a class C. This is called [Class Conditional Probability]

Ex: If there are two classes C_1 and C_2

$P(C_1/x) > P(C_2/x)$

Then x belongs to C_1

$P(C_1/x) = \frac{P(C_1) \cdot P(x/C_1)}{P(x)}$

$P(C_2/x) = \frac{P(C_2) \cdot P(x/C_2)}{P(x)}$

find ⇒ $\boxed{P(C_1) \cdot P(x/C_1) > P(C_2) \cdot P(x/C_2)}$

if and allot class

Ex: Tumor size: Big/Small

age: Young/old/Child

Class: - B/M → (Benign/Malig)

	Size	Age	Class
1	Big	Y	B
2	Small	Old	B
3	Small	Y	B
4	Big	Old	B
5	Big	C	M
6	Small	O	B
7	Big	C	B
8	Small	Y	M
9	Small	O	B
10	Small	Y	M

⇒ $P(B) = \frac{7}{10}$

⇒ $P(M) = \frac{3}{10}$

For Multiple features: [All features are independent]

⇒ eg:- Tumor type (feature 1) differ from Tumor size (feature 2)

$x = (x_1, x_2 \dots x_n)$

$P(C/x_1, x_2 \dots x_n) \approx P(C) \cdot P(x_1, x_2 \dots x_n / C)$

$\approx P(C) \cdot P(x_1/C)\, P(x_2/C) \dots P(x_n/C)$

$\approx P(C) \prod_{i=1}^{n} P(x_i/C)$

$x = \{Big, Young\}$

$P(B/Big, Young)$

$= P(B) \cdot P(Big, Y/B)$

$P(M/Big, Young)$

$= P(M) \cdot P(Big, Y/M)$

$P(Big, Young \mid B) \Rightarrow P(Big \mid B) \cdot P(Young \mid B)$

$\Rightarrow \left(\frac{3}{7}\right)\left(\frac{2}{7}\right) = \frac{6}{49}$

$P(Big, Young \mid M) \Rightarrow \left(\frac{1}{3}\right)\left(\frac{2}{3}\right) = \frac{2}{9}$

$P(B \mid Big, Young) = P(B) \cdot P(Big, Young \mid B)$

$= (0.7)\left(\frac{6}{49}\right)$

$= \frac{6}{70} = \frac{1}{11.6}$

$P(M \mid Big, Young) = P(M) \cdot P(Big, Young \mid M)$

$= (0.3)\left(\frac{2}{9}\right)$

$= \frac{2}{30}$

$= \frac{1}{15}$

$P(B \mid Big, Young) > P(M \mid Big, Young)$

Given New ex, x:

$$P(Y=1 \mid x) = \frac{P(x \mid Y=1)\, P(Y=1)}{P(x)}$$

$$P(x) = \sum_{y \in Y} P(x, Y)$$

$$\boxed{P(x) = P(x \mid Y=0) \cdot P(Y=0) + P(x \mid Y=1) \cdot P(Y=1)}$$

Let, $P(x \mid Y=0) = 0.01$, $P(x \mid Y=1) = 0.03$, $P(Y=1) = P(Y=0) = 0.5$

$$P(Y=1 \mid x) = \frac{P(x \mid Y=1)\, P(Y=1)}{P(x)} = \frac{(0.03) \cdot 0.5}{0.020}$$

$$P(x) = (0.01)(0.5) + (0.03)(0.5)$$

$$= 0.005 + 0.015$$

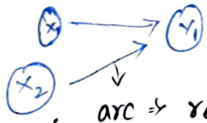
$$= 0.020$$

$$\frac{0.015}{0.020}$$

$$= \frac{15}{20}$$

$$= \frac{3}{4} \Rightarrow \boxed{0.75}$$

Bayesian Belief Network: features are dependent here [There is some Relation b/w Variables]



arc ⇒ represents relation b/w variables

⇒ No arc ⇒ No relation.

⇒ This is directed Acyclic Graph

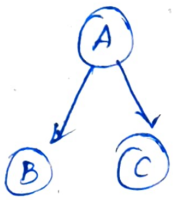
⇒ x is parent

⇒ y is child

$$P(x_1 \ldots x_n) = \prod_{j=1}^{n} P(x_j \mid Parent(x_j))$$

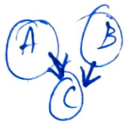
Ex:

Case. 1:



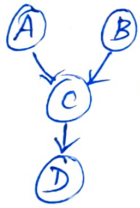
$$\Rightarrow P(A,B,C) = [P(B|A)][P(C|A)][P(A)]$$

Case 2:



$$\Rightarrow P(A,B,C) = P(C|A,B) \cdot P(A) \cdot P(B)$$

Case 3:



$$\Rightarrow P(A,B,C,D) = P(D|C) \cdot P(C|A,B) \cdot P(A) \cdot P(B).$$

Decision Tree:

Decision 1

⇒ How to choose what feature to split on at each node? At the root node as well as on the left branch or right branch of the decision tree

Ex:

Age, Tumor Size, Color → which feature result in the greatest purity of the labels on the left and right substances.

Highly pure subset of examples < predict 1, predict 0

Maximum Purity ⟶ entropy fn ⇒ [Estimates Impurity]

DECISION - 2

How do you stop splitting? (i) when a node is 100% of one particular class

(*) It may take a lot of time

(ii) Maximum depth → fix the parameter.

Depth: → In a DT, the depth of a node is defined as the no. of hops one take to get from root node.

→ root node is at "zero depth" (or) "depth zero" as it takes zero hops to get to itself.

Entropy: Impurity of the set of

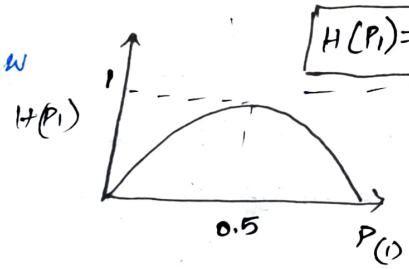
Ex: [1] [1] [1] [0] [0] [0] → from one node,

Let P_1 = fraction of examples which are in class 1

for (1) class → $P_1 = \frac{1}{2}$

$H(P_1) = 1$ → highly impure

$$\boxed{H(P_1) = -P_1 \log_2(P_1) - P_0 \log_2(P_0)} \rightarrow \boxed{P_0 = 1 - P_1}$$



→ If 50 - 50, curve is highest

→ when all in 1 → $H(P_1) = 0$

all in 0 → $H(P_1) = 0$

→ as $P_1 = 1$ for all classes.

[only one class present at a node]

Ex: 0, 1, 1, 1, 1, 1

$P_1 = \frac{5}{6} = 0.83$, $\boxed{H(P_1) = 0.65}$

∴ If $P_1 = 0$ or $P_0 = 0$

$0 \log_2(0) = 0$ by convention

Information: When building a DT, → what feature to split on

(i) based on which choice of feature to reduce the entropy most

(ii) Reduce entropy (or) maximize purity

(iii) Reduction of entropy → Information gain

Ex:

Age

old → 1 0 1 1 1

young → 0 1 0 0 0

$P_1 = \frac{4}{5} = 0.8$

$\boxed{H(P_1) = 0.72}$

$P_1 = \frac{1}{5} = 0.2$

$H(P_1) = [-0.2(\log_2(0.2)) - (0.8)\log_2(0.8)]$

$+ 0.46 + 0.257$

$= +0.717$

$\boxed{H(P_1) = 0.72}$

Ex: To split on Tumor shape

Tumor shape

R → 0, 1, 1, 1, 0, 0, 1

NR → 1, 0, 0

$P_1 = \frac{4}{7} = 0.57$

$P_1 = \frac{1}{3} = 0.33$

$\Rightarrow [+0.52 + 0.38]$

$= -[0.57(\log_2(0.57) + 0.43\log_2(0.43)] \Rightarrow 0.92$

$= -[-0.46 - 0.52]$

$= 0.98$

3) Case III :

Color

Light

1, 1, 0, 1

$P_1 = 0.75$

$H(P_1) = 0.83$

Dark

0, 1, 0, 1, 0, 0

$P(1) = 0.33$

$H(0.33) = 0.92$

To find which feature works best, use weighted average

not used now..

Weighted Average:

Age $\Rightarrow \left(\frac{5}{10} H(0.8) + \frac{5}{10} H(0.2) \right)$

~~Total~~ no. of data / Total data

5 datas present in old / 10 ⇒ total data

$\Rightarrow \frac{5}{10}[0.72] + \frac{5}{10}[0.72]$

$\Rightarrow 0.72$

For Tumor shape,

$\frac{7}{10} H(0.57) + \frac{3}{10} H(0.33)$

$\Rightarrow 0.686 + 0.276$

$\Rightarrow 0.962$

For colour: $\frac{4}{10} H(0.75) + \frac{6}{10} H(0.33)$

$\Rightarrow \frac{4}{10}(0.83) + \frac{6}{10}(0.92)$

$\Rightarrow 0.332 + 0.552$

$\Rightarrow 0.884$

Lesion Tree:

Age	Tumor shape	Color	Class	Weight
Old	R	Light	1	7.2
Young	NR	Light	1	8.8
Young	R	Dark	0	15
Old	NR	Light	0	9.2
Old	R	Light	1	8.4
Old	R	Dark	1	7.6
Young	NR	Dark	0	11
Old	R	Dark	1	10.2
Young	R	Dark	0	18
Young	R	Dark	0	20

$H(P_1) = -P_1 \log_2(P_1) - P_0 \log_2(P_0)$

$\Rightarrow P_1 \log_2$

$\Rightarrow -P_1 \log_2 P_1 - P_0 \log_2(P_0)$

Q] How to choose what feature to split on at each node ?

Age

Old → 10111 → $H(OB) \Rightarrow 0.72$

Young → 10000 → $P_1 = \frac{1}{5} \Rightarrow 0.2$, $H(A) \Rightarrow 0.72$

Weighted Average: "Refer prev page"

Age ⇒ 0.72 , Tumor shape ⇒ 0.962

Color ⇒ 0.884

Change in Rule: (i) Compute reduction in entropy compare to if we had not split it all.

(ii) For doing this, before splitting find H()

Information Gain / Reduction in entropy:

Before Splitting:

$\frac{5}{10} = P_1 = 0.5$

$-0.5 \log_2(0.5) - 0.5 \log_2(0.5)$

$= (0.5)(-1) - (0.5)(-1)$

$\boxed{H() \Rightarrow 1}$ → H(0.5) always 1 → highest entropy

Reduction in entropy ⇒ Before splitting − Weighted average

(i) H(0.5) − Weighted Average of Age

⇒ 1 − 0.72

⇒ 0.28

(ii) H(0.5) − W_A of Tumor shape

⇒ 1 − 0.962

⇒ 0.038

(iii) H(0.5) − W_A of Color

⇒ 1 − 0.884

⇒ 0.116

~~In color~~

⇒ Biggest Reduction in entropy is in "Age."

⇒ Then split on Age. Same Rule is followed recursively for further splitting, until stopping Criteria is met.

⇒ when a node is 100% pure,

⇒ when splitting a node will result in the tree exceeding a max depth.

⇒ Information gain from additional splits is less than threshold.

⇒ when no. of examples in a node below a threshold.

Age

old — Young

Tumor shape — Color

R — NR

1 — 0

1

1

1

$H(P_1) \Rightarrow 0$

Regression Problem:

Age

Old — Young

Tumor shape — NR

Weights: R: 7.2, 8.4, 8.6, 10.2

Average of weight if R

NR Weights: ~~15, 18, 20~~ 9.2,

Tumor shape

Same process of Average weight

⇒ "Variation of weight should be less"

Eg:

Age

old — Young

7.2, 9.2, 8.4, 7.6, 10.2 — Variance : 1.47

8.8, 15, 18, 18, 20 — Variance : 21.87

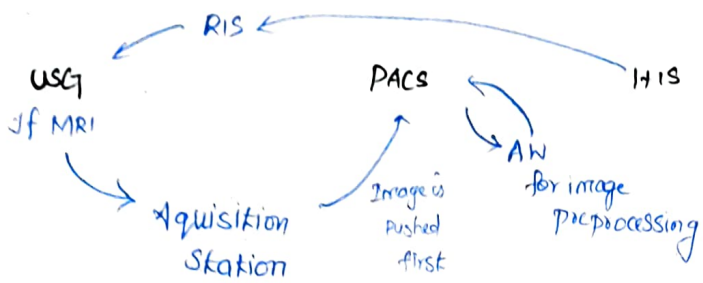
⇒ Variance before splitting ⇒ 20.5.

Modification of Rule: Reduction in Variance.

⇒ Variance before splitting $- \left[\frac{5}{10} \times 1.47 + \frac{5}{10} \times 21.87\right]$

⇒ 8.84

PACS



∴ CT has -2000 to 2000 Hounsfield units in hospital setup
1000 μU

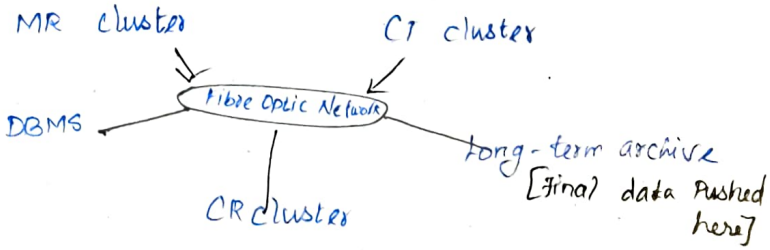
→ MOST radiological images / videos has 10 bit per pixel
∴ normally 8 bit for gray scale.

PACS architecture:

LAN, WAN. In PACS instead of WAN, Telemedicine Port is used.
→ No sensitive info transferred via Telemedicine Port
↳ exclusions: patient ~~Radiological~~ Surgical video data etc.
→ Patient Surgical video pushed to PACS via LAN where annotations and post processing alone that is pushed back

Centralized PACS: Short term storage.
But does not work in big hospitals.

Distributed PACS:
⇒ developed by Mayo Clinic
⇒ Runs 65 MRI Scan flows across US
⇒



→ Image servers from the clusters are connected through Fibre Optic Network ~~then~~.

Disadvantages:
→ Expensive, Complex Tech.
→ Maintenance Programme required.
→ Requirement of dedicated Personnel
→ No fall back position conventional with

DICOM: Digital Imaging and Communication in Medicine.

↳ standard for handling, storing, printing and transmitting information on medical imaging.

Why DICOM →

↳ Workflow management. incl. preparation of worklists [sequential]

↳ Indication given to doctors about quality of images.

∴ JPEG → uses wavelet for processing!

∴ Psychodelic Compression Mechanism used in hearing aids! → Same in JPEG.

[300-8k Hz is seperated]

↓

→ Images seperated into high and low freq components [May be using fourier]

→ a wavelet is used to get low freq amplified that contains structural data and supress high freq (noise).

Decision Tree

Age	Vaccination	Tumor Size	Tumor Site	Malignant
5	1	Small	Shoulder	0
9	1	Small	Knee	1
6	0	Small	Marrow	0
6	1	Medium	Chest	0
7	0	M	Shoulder	0
8	1	L	Shoulder	0
5	1	L	Liver	0
9	0	S	Liver	1
8	0	M	Shoulder	1
8	0	M	Shoulder	1
6	0	S	Marrow	1
7	0	S	Chest	1

(i) What is the initial entropy of Malignant?

$H(P_1) = -P_1 \log_2(P_1) - (1-P_1)\log_2(1-P_1)$

Vaccination

→ 1 Malig: 0 0 0 0 0

→ 0 Malig: 0 0 1 1 1 1 1

$P_1 = \frac{5}{7}$

$H(P_1) = -\frac{5}{7}\log_2\left(\frac{5}{7}\right)$

$\frac{2}{7}\log_2\left(\frac{2}{7}\right)$

$= 0.71(2.82)$

$= -0.71(-0.49) - 0.28(-1.83)$

$= 0.3479 + 0.5124$

$H\left(\frac{5}{7}\right) = 0.86$

(ii) Information gain:

$\Rightarrow 0.9798 - \left\{\frac{5}{12} H(0) + \frac{7}{12} H\left(\frac{5}{7}\right)\right\}$

$\Rightarrow 0.9798 - \left[\frac{7}{12}[0.86]\right]$

$\Rightarrow 0.9798 - 0.501$

$\Rightarrow 0.4788$

Info gain:

$H(P_1 \text{ root}) - \left[\frac{10}{12}\text{left } H(P_1 \text{ left}) + \frac{10}{12}\text{right } H(P_1 \text{ right})\right]$

$= -\frac{5}{12}\log\left(\frac{5}{10}\right) - \left(\frac{7}{12}\right)\log_2\left(\frac{7}{10}\right)$

0.9798

(iv) Before split, group of 5 cats and 5 non cats entropy $H(5/10) = H(0.5)$

1 1 1 1 0 0 0 | 1 0 0

Info gain $\Rightarrow H(0.5) - \left\{\frac{7}{10} H\left\{\frac{4}{7}\right\} + \frac{3}{10} H\left\{\frac{1}{3}\right\}\right\}$

before splitting

$= 1 - \left\{\frac{7}{10}\left[\left(\frac{4}{7}\right)\log_2\left(\frac{4}{7}\right) - \left(\frac{3}{7}\right)\log_2\left(\frac{3}{7}\right)\right] + \left(\frac{3}{10}\right)\left(-\frac{1}{3}\log_2\left(\frac{1}{3}\right) - \frac{2}{3}\log_2\left(\frac{2}{3}\right)\right)\right\}$

$= 1 - \left[0.7\left(-(0.571)(0.80) - (0.429)(1.36)\right) + (0.3)\right]$

~~Rule~~ are the encoding of categorical features.

Tumor shape-R	Tumor shape-NR	Tumor shape-Oval
1	0	0
0	1	0
0	0	1

} one hot encoding

~~Rule: categorical → k binary features~~

Rule: categorical → k possible → construct k binary features that only can take values 0 or 1.

[Decision tree is also called interpretable model]

because small decision trees can be interpreted easily [Rules can be interpreted]

∴ [Random Forest is ensembled decision trees]

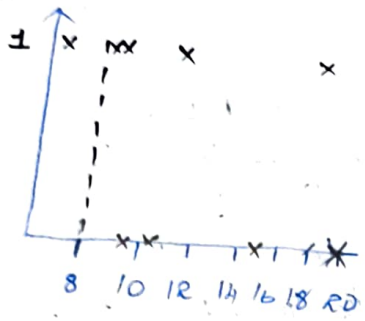
Age	Tumor shape	Color	wt	class
			7.2	1
			8.8	1
			15	0
			9.2	0
			8.4	1
			11	1
			10.2	0
			18	1
			20	0

→ Sort these values and there will be 9 midpoints totally. These points can be taken as threshold.

→ Then split it and calculate info gain.

→ High info gain threshold is chosen.

* midpoints are b/w two points.



check whether $w^* \le 8$ → random (wt)

[1 1] | 1 0 0 1 0 1 0 0

$H(2/2) = 0$ $H(3/8)$

info gain ⇒ for $w^* \le 8$

$$H(0.5) - \left(\frac{2}{10}(H(0)) + \frac{8}{10} H(3/8)\right)$$

$= 0.24$

Eigen Vectors.

* What will happen when a matrix hits a vector?

let $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ $\left(x = \begin{bmatrix} 1 \\ 3 \end{bmatrix} \right.$ $Ax = \begin{bmatrix} 7 \\ 5 \end{bmatrix}$

It changes path ie, it transforms into new vector.

⇒ But in eigens, it doesn't transform.

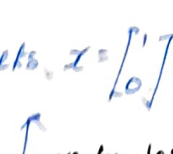
⇒ for every square matrix A, there exits a special vector, which is its eig

$Ax = \lambda x$ $x \Rightarrow$ eigen vector, $\lambda \Rightarrow$ eigen value.

Basis of a Matrix: A set of vectors $\in \mathbb{R}^n$ is called a basis, if they are linearly independent and every vector in $\mathbb{R}^n$ can be expressed as a linear combination of these vectors.

Linearly Independent: A set of n vectors are linearly independent if no vector in the set can be expressed as a linear combination of that remaining $(n-1)$ vectors.

Lets $x = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $y = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$



$\Downarrow$

$\begin{bmatrix} a \\ b \end{bmatrix} = a \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

linearly independent

$c_1 v_1 + c_2 v_2 + \dots c_n v_n = 0$, where all c_i are scalars

⇒ If this eq^n has zero solution, $\{v_1, v_2 .. v_n\}$ are linearly independent set of vectors.

⇒ In general, any linearly independent set of vectors with n elements is a basis of $\mathbb{R}^n$.

$c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ are basis of $\mathbb{R}^2$

Ex: $\mathbb{R}^2$, let $v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $v_2 = \begin{bmatrix} 3 \\ -5 \end{bmatrix}$

$c_1 v_1 + c_2 v_2 = 0$

$\Rightarrow c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 3 \\ -5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$\Rightarrow c_1 + 3c_2 = 0$

$2c_1 - 5c_2 = 0$

$-6c_2 - 5c_2 = 0$

$c_2 = 0$, $c_1 = 0$

Two goals of PCA: [from the previous table].

⇒ Along these dimensions (x and y), data has high variance

⇒ The dimension should be uncorrelated. [linearly Independent]

⇒ Even better if these new dimensions are orthogonal.

⇒ **With a New basis:**

⇒ Let $V_1, V_2 \ldots V_n$ be a set of such n linearly independent orthonormal vectors.

⇒ Let P be an $n \times n$ matrix such that $V_1, V_2 \ldots V_n$ are columns of P.

$$P = \begin{bmatrix} \uparrow & \uparrow & & \uparrow \\ V_1 & V_2 & \cdots & V_n \\ \downarrow & \downarrow & & \downarrow \end{bmatrix}$$

⇒ Let $x_1, x_2, \ldots, x_m \in \mathbb{R}^n$ be m data points and let X be a matrix such that $x_1, x_2 \ldots x_m$ are the rows of the matrix. Assume the data has "0 mean"

making the data 0 mean is Standardization. (Scaling process).

when

$$X = \begin{array}{c} x_1 \rightarrow \\ x_2 \rightarrow \\ \vdots \\ x_m \rightarrow \end{array} \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \\ & \\ x_{m1} & \end{bmatrix} \in \mathbb{R}^n \Rightarrow \begin{bmatrix} x_{11} & \cdots \\ x_{21} & \cdots \\ \vdots & \\ x_{m1} & \end{bmatrix}$$

mean μ_1

⇒ New scaled $X = \begin{bmatrix} x_{11} - \mu_1 & \cdots \\ x_{21} - \mu_1 & \cdots \\ \vdots & \\ x_{m1} - \mu_1 & \cdots \end{bmatrix}$

Similar process of all columns

$$\begin{bmatrix} x_{21} - \mu_2 & \cdots \\ x_{22} - \mu_2 & \cdots \\ \vdots & \\ x_{2n} - \mu_2 & \cdots \end{bmatrix}$$

⇒ The data with zero mean n,

$$X = \begin{array}{c} x_1 \rightarrow \\ x_2 \rightarrow \\ \vdots \\ x_m \rightarrow \end{array} \begin{bmatrix} f_1 & f_2 & \cdots & f_n \end{bmatrix} \in \mathbb{R}^n$$

$m \times n$

x_i vector.

⇒ Each x_i is represented using the new basis P.

$$x_i = c_{i1} V_1 + c_{i2} V_2 + \cdots c_{in} V_n$$

$$c_{i1} = x_i^T V_1$$
$$c_{i2} = x_i^T V_2$$
$$\vdots$$
$$c_{in} = x_i^T V_n$$

⇒ [Representing data with new dimensions $c_1, c_2 \ldots c_i$]

c_{i1}

$\hat{x}_i$

⇓

This opera

⇓

$X = \ldots$

⇒ **Theorem:** If X has zero mean. Then

where to get P matrix:

↳ Covariance Matrix: $\Sigma = \ldots$

Covariance formula

⇒ The dimensio

⇒ Let there be

⇒ Let c be the the means of

$$c_{i_1} = \begin{bmatrix} \leftarrow & x_i^T & \rightarrow \end{bmatrix} \begin{bmatrix} \uparrow \\ v_i \\ \downarrow \end{bmatrix}$$

$$\hat{x}_i = \begin{bmatrix} \leftarrow & x_i^T & \rightarrow \end{bmatrix} \begin{bmatrix} \uparrow & \uparrow & & \uparrow \\ v_1 & v_2 & \cdots & v_n \\ \downarrow & \downarrow & & \downarrow \end{bmatrix}$$ ⇒ single row vector formed $1 \times n$.

⇓

This operation is done to all the x_i in the X matrix.

⇓

$$X = \begin{bmatrix} \leftarrow & x_1 & \rightarrow \\ \leftarrow & x_2 & \rightarrow \\ & \vdots & \\ \leftarrow & x_n & \rightarrow \end{bmatrix} \begin{bmatrix} \uparrow & \uparrow & & \uparrow \\ v_1 & v_2 & \cdots & v_n \\ \downarrow & \downarrow & & \downarrow \end{bmatrix}$$

$m \times n$ $\quad$ $n \times m$ $\quad$ P matrix

⇒ matrix with $m \times m$ matrix.

⇒ <u>Theorem:</u> If x is a matrix such that all ~~the~~ its column has zero mean, If $\hat{x} = xP$, then column of $\hat{x}$ has "zero mean."

<u>Where to get P matrix??</u>

↳ <u>Covariance Matrix:</u> Let X is a matrix whose columns are zero mean. Then $\Sigma = \frac{1}{m} X^T X$ is the co-variance matrix.

<u>Covariance</u> formula ⇒ $$P_{yz} = \frac{\sum_{i=1}^{m} (y_i - \bar{y})(z_i - \bar{z})}{\sqrt{\sum (y_i - \bar{y})^2} \sqrt{\sum (z_i - \bar{z})^2}}$$

⇒ The dimension of P_{yz} is $n \times n$.

⇒ Let there be i and j columns.

$$\begin{array}{ccc} & i & j \\ x_1 & x_{1i} & x_{1j} \\ x_2 & x_{2i} & x_{2j} \\ \vdots & \vdots & \vdots \\ x_m & x_{mi} & x_{mj} \end{array}$$

⇒ Let C be the covariance matrix of X. Let μ_i and μ_j be the means of ith column and j^{th} column

$$C_{ij} = \frac{1}{m} \sum_{k=1}^{m} [x_{ki} - \mu_i][x_{kj} - \mu_j]$$

$$= \frac{1}{m} \sum_{k=1}^{m} x_{ki} x_{kj}$$

$$= \frac{1}{m} x_i^T x_j$$

$$\cancel{= \frac{1}{m} (x^T x)_{ij}}$$

$$= \frac{1}{m} [x^T x]_{ij}$$

For a covariance matrix:

$$\Sigma = \begin{bmatrix} Var(x) & cov(x,y) \\ cov(y,x) & Var(y) \end{bmatrix}$$

covariance matrix

→ ideally $cov(x,y)$ and $cov(y,x)$ should be zero. for the choosen dimension.

⇒ Σ is a square ~~matrix~~ symmetric matrix.

⇒ eigen vectors of square symmetric matrix are orthogonal.